

SECTION B

7 (**long**) A solid sphere of mass m and radius r is rolling down a slope whose angle to the horizontal is α as shown in Fig. 4. The velocity of the sphere is denoted v and its angular velocity is ω .

- (a) Find an expression for the angular acceleration $\dot{\omega}$ in terms of the acceleration $\dot{v}$ in the case where the sphere rolls without slip. [4]
- (b) Show that the normal reaction N between the sphere and the table is $N = mg \cos \alpha$. [4]
- (c) Using d'Alembert's principle or otherwise find an expression in terms of α for the minimum coefficient of friction μ required for the sphere to roll without slip. For the case $\alpha = \pi/4$ show that $\mu = \frac{2}{7}$. [15]
- (d) For the case $\alpha = \pi/4$ and the coefficient of friction is $1/7$, calculate $\dot{\omega}$ and $\dot{v}$. [7]

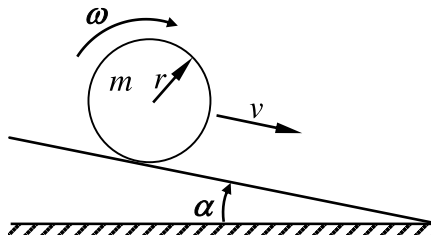


Fig. 4